

Session 05 (AIML) -Assignment Question

- **Name of Student :** Shruti Dnyaneshwar Darwatkar
- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
1  # Take string input from user
2  s = input("Enter a string: ")
3
4  # Positive indexing starts from 0
5  print("First character:", s[0])
6
7  # Negative indexing starts from -1
8  print("Last character:", s[-1])
9
10 print("3rd character from start:", s[2])
11
12 print("2nd character from end:", s[-2])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Microsoft\WindowsApps\python3.9.0-win32-x64\python.exe' s-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy
Enter a string: shruti
First character: s
Last character: i
3rd character from start: r
2nd character from end: t
PS C:\Users\tanve\Desktop\ITR> s
```

Q2.

PROGRAM:

```
session 5.py > ...  
1  # Take string input from user  
2  s = input("Enter a string: ")  
3  
4  # Slicing syntax: string[start:end:step]  
5  
6  print("First 5 characters:", s[:5])  
7  
8  print("Last 4 characters:", s[-4:])  
9  
10 print("String in reverse order:", s[::-1])  
11  
12 print("Every 2nd character:", s[::2])
```

OUTPUT:

```
op\ITR\session 5.py  
First 5 characters: shrut  
Last 4 characters: ruti  
String in reverse order: iturhs  
Every 2nd character: srt  
PS C:\Users\tanve\Desktop\ITR>
```

Q3.

PROGRAM:

```
session 5.py > ...
1  # Take main string and substring input
2  main_str = input("Enter the main string: ")
3  sub_str = input("Enter the substring: ")
4
5  # Check membership using 'in'
6  if sub_str in main_str:
7      print("Substring found!")
8  else:
9      print("Substring not found!")
10
11 # Check membership using 'not in'
12 if sub_str not in main_str:
13     print("Substring not found!")
```

OUTPUT:

```
op\ITR\session 5.py'
Enter the main string: shruti darwatkar
Enter the substring: shruti
Substring found!
PS C:\Users\tanve\Desktop\ITR>
eted.
```

Q4.

PROGRAM:

```
session 5.py > ...
1  # Take string input
2  s = input("Enter a string: ")
3
4  # Length of string
5  length = len(s)
6
7  print("Length of string:", length)
8
9  # Last character using len()
10 print("Last character:", s[length - 1])
11
12 # Middle character if length is odd
13 if length % 2 != 0:
14     print("Middle character:", s[length // 2])
15 else:
16     print("String length is even, no single middle character.")
17
18 # Common mistake:
19 # Last index = len(s) - 1
20 # Using s[len(s)] causes IndexError
```

OUTPUT:

```
op\ITR\session 5.py
Enter a string: shruti
Length of string: 6
Last character: i
String length is even, no single middle character.
PS C:\Users\tanve\Desktop\ITR>
```

Q5.

PROGRAM:

```
session 5.py > ...
1  # range(stop)
2  print("Numbers from 0 to 10:")
3  for i in range(11):
4      print(i)
5
6  # range(start, stop)
7  print("\nNumbers from 5 to 15:")
8  for i in range(5, 16):
9      print(i)
10
11 # range(start, stop, step)
12 print("\nOdd numbers from 1 to 21:")
13 for i in range(1, 22, 2):
14     print(i)
15
16 # Forms of range():
17 # range(stop)
18 # range(start, stop)
19 # range(start, stop, step)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\t  
'c:\Users\tanve\.vscode\extensions\ms-python.debu  
op\ITR\session 5.py'
```

```
2  
3  
4  
5  
6  
7  
8  
9  
10
```

```
Numbers from 5 to 15:
```

```
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15
```

```
Odd numbers from 1 to 21:
```

```
1  
3  
5  
7  
9  
11  
13  
15  
17  
19  
21
```

```
PS C:\Users\tanve\Desktop\ITR> 
```

Q6.

PROGRAM:

```
session 5.py > ...
1  # Numbers from 20 down to 10
2  print("Numbers from 20 down to 10:")
3  for i in range(20, 9, -1):
4      print(i, end=" ")
5
6  print("\n")
7
8  # Numbers from 15 down to 1 in steps of 2
9  print("Numbers from 15 down to 1 in steps of 2:")
10 for i in range(15, 0, -2):
11     print(i, end=" ")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR\session 5.py'
Numbers from 20 down to 10:
20 19 18 17 16 15 14 13 12 11 10

Numbers from 15 down to 1 in steps of 2:
15 13 11 9 7 5 3 1
PS C:\Users\tanve\Desktop\ITR> 
```


Q7.

PROGRAM:

```
session 5.py > ...
1  # Take string input
2  s = input("Enter a string: ")
3
4  print("Characters with their indices:")
5  for i in range(len(s)):
6      print(i, ":", s[i])
7
8  print("\nString in reverse order:")
9  for i in range(len(s)-1, -1, -1):
10     print(s[i], end="")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR\session 5.py'
Characters with their indices:
0 : s
1 : h
2 : r
3 : u
4 : t
5 : i

String in reverse order:
iturhs
PS C:\Users\tanve\Desktop\ITR> 
```


Q8.

PROGRAM:

```
session 5.py > ...
1  # Take number input
2  num = int(input("Enter a number: "))
3
4  # Check in range(1, 51)
5  if num in range(1, 51):
6      print(num, "is present in range(1, 51)")
7  else:
8      print(num, "is NOT present in range(1, 51)")
9
10 # Check in range(10, 100, 5)
11 if num in range(10, 100, 5):
12     print(num, "is present in range(10, 100, 5)")
13 else:
14     print(num, "is NOT present in range(10, 100, 5)")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\vscode\extensions\ms-python\ITR\session 5.py'
Enter a number: 26
26 is present in range(1, 51)
26 is NOT present in range(10, 100, 5)
PS C:\Users\tanve\Desktop\ITR>
```

Q9.

PROGRAM:

```
session 5.py > ...
1  # First 10 even numbers (2 to 20)
2  print("First 10 even numbers:")
3  for i in range(2, 21, 2):
4      print(i, end=" ")
5
6  print("\n")
7
8  # Numbers from 1 to 30 in steps of 3
9  print("Numbers from 1 to 30 in steps of 3:")
10 for i in range(1, 31, 3):
11     print(i, end=" ")
12
13 print("\n")
14
15 # String slicing
16 s = "PythonProgramming"
17
18 print("Python :", s[:6])
19 print("Programming :", s[6:])
20 print("Reverse :", s[::-1])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> cd c:\Users\tanve\Desktop\ITR
'c:\Users\tanve\.vscode\extensions\ms-python.python-2019.10.2\python.exe'
op\ITR\session 5.py
First 10 even numbers:
2 4 6 8 10 12 14 16 18 20

Numbers from 1 to 30 in steps of 3:
1 4 7 10 13 16 19 22 25 28

Python : Python
Programming : Programming
Reverse : gnimmargorPnohtyP
PS C:\Users\tanve\Desktop\ITR>
```

Q10.

PROGRAM:

```
session 5.py > ...
1  # String Analyzer Program
2
3  s = input("Enter a string: ")
4
5  # 1. Length of string
6  print("Length of string:", len(s))
7
8  # 2. First half and second half
9  mid = len(s) // 2
10
11 print("First Half:", s[:mid])
12 print("Second Half:", s[mid:])
13
14 # 3. Check if 'python' exists (case-insensitive)
15 if "python" in s.lower():
16     print("'python' is present in the string.")
17 else:
18     print("'python' is not present in the string.")
19
20 # 4. Print characters with positive and negative indices
21 print("\nCharacter\tPositive Index\tNegative Index")
22 for i in range(len(s)):
23     print(s[i], "\t\t", i, "\t\t", i - len(s))
24
25 # 5. Reverse string
26 print("\nReverse String:", s[::-1])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\vscode\extensions\ms-python.debugop\ITR\session 5.py'
Enter a string: shruti
Length of string: 6
First Half: shr
Second Half: uti
'python' is not present in the string.

Character      Positive Index  Negative Index
s              0             -6
h              1             -5
r              2             -4
u              3             -3
t              4             -2
i              5             -1

Reverse String: iturhs
PS C:\Users\tanve\Desktop\ITR> 
```